

HARDIK VIKAS JAIN

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EDUCATION

Vellore Institute of Technology (VIT), Vellore, Tamil Nadu

Bachelor of Technology in Computer Science and Engineering (Information Security) July 2024 – May 2028

CGPA : 9.15/10.00

TECHNICAL SKILLS

- Languages: C, C++, Python, JavaScript, SQL, HTML5, CSS3
- Systems & Backend: Linux Kernel Primitives, FastAPI, Node.js, Socket Programming
- Databases & Infrastructure: PostgreSQL, SQLite, Oracle SQL, Docker, Kubernetes, Git, GitHub, Linux, VS Code
- Core Computer Science: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Object-Oriented Programming (OOP), Compiler Design
- AI & Data Science: Hybrid RAG, BM25, Vector Embeddings, Reciprocal Rank Fusion (RRF), Cross-Encoders, Pandas, NumPy, Generative AI

TECHNICAL PROJECTS

Enterprise Hybrid RAG Platform | *FastAPI, Docker, Kubernetes, BM25, Dense Vector Search, RRF*

- Architected a production-ready Retrieval-Augmented Generation (RAG) microservice using FastAPI, integrating sliding-window chunking algorithms and automated pipeline evaluation harnesses.
- Engineered a high-precision hybrid retrieval system combining BM25 sparse keyword matching with dense vector embeddings, ranked via Reciprocal Rank Fusion (RRF) and cross-encoder re-ranking.
- Containerized end-to-end workflows via Docker & Kubernetes manifests; integrated token bucket rate-limiting algorithms and API-key security guards to mitigate Denial-of-Service vectors.

LAN Watchtower — Network Telemetry & Host Discovery | *Python, Scapy, CustomTkinter, SQLite*

- Developed an asynchronous network monitoring desktop tool executing high-speed, multi-threaded ARP/ICMP sweeps across complex CIDR subnet ranges.
- Automated targeted TCP port vulnerability inspections across key services (HTTP, SSH, SMB, FTP) with transactional SQLite WAL (Write-Ahead Logging) telemetry storage and real-time host discovery alerts.

Personal Developer Portfolio Website | *HTML5, CSS3, JavaScript, Git, Vercel CI/CD*

- Designed and developed an interactive, responsive developer portfolio utilizing core web technologies, ensuring cross-browser compatibility and an optimal user experience.
- Achieved a 98+ Google Lighthouse score across performance, accessibility, and SEO metrics by utilizing modern responsive paradigms and asset minification.
- Configured Git-triggered Vercel CI/CD pipelines to ensure automated, zero-downtime deployments.

End-to-End Custom Compiler | *C, Flex, Bison, Assembly, Data Structures*

- Architected a full-pipeline, multi-pass compiler translating a custom high-level C-subset into executable Assembly, comprehensively executing all six standard compilation phases: Lexical Analysis, Syntax Analysis, Semantic Analysis, Intermediate Code Generation, Code Optimization, and Target Code Generation.
- Engineered a high-precision frontend combining a Flex-based lexical scanner and a Bison LALR(1) parser to construct Abstract Syntax Trees (AST) and enforce strict semantic type-checking via dynamically managed symbol tables.
- Optimized execution pipelines by generating machine-independent Three-Address Code (TAC), applying foundational dead-code elimination techniques, and mapping the intermediate representation directly into target machine instructions.

ACHIEVEMENTS & ACTIVITIES

- VIT Code to Innovate Hackathon: Participant and Technical Contributor, designing modular system architectures under competitive sprint timelines.
- 1st Position — CIVIL HUNT: Awarded first place in the flagship competitive problem-solving event at the RIVERA Cultural & Sports Fest.
- Relevant Coursework: Advanced Data Structures & Algorithms, Computer Networks, Operating Systems, Database Management Systems, Applied Cryptography.

INTERESTS & HOBBIES

- Building Scalable Software Projects and Exploring Generative Artificial Intelligence Architectures.
- Solving Complex Data Structures & Algorithms Problems and Competitive Programming.